Discovering Novel *Mycobacterium tuberculosis* (M.tb) mutations

Supervisors: Dr. Michael Hall, Prof. Lachlan Coin

Nabil Islam

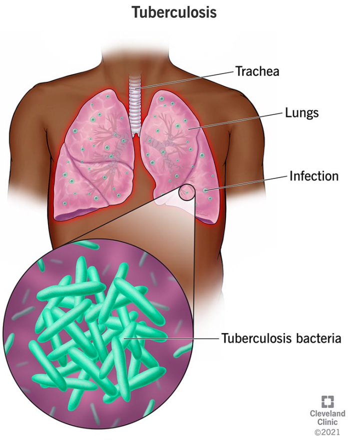
nabil.islam@student.unimelb.edu.au

THE UNIVERSITY OF MELBOURNE

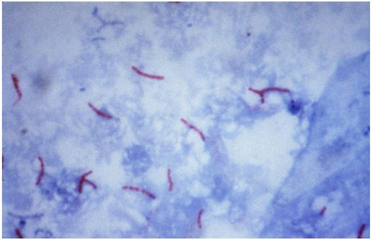
Department of Microbiology and Immunology
The University of Melbourne

May 24, 2026

What is Tuberculosis(TB)?



Cleveland Clinic

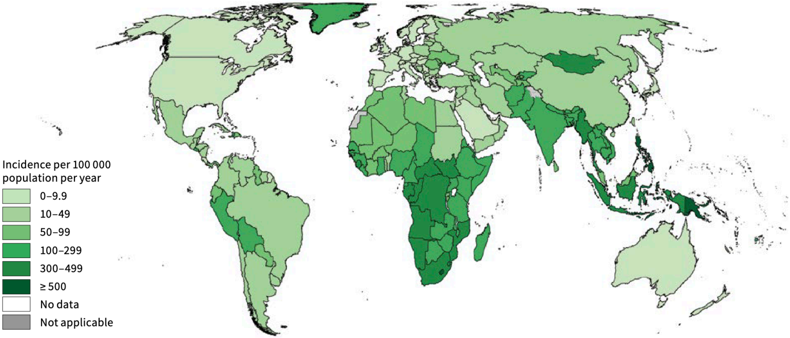


Centers for Disease Control and Prevention

The burden of TB

FIG. 5

Estimated TB incidence rates at country level, 2024

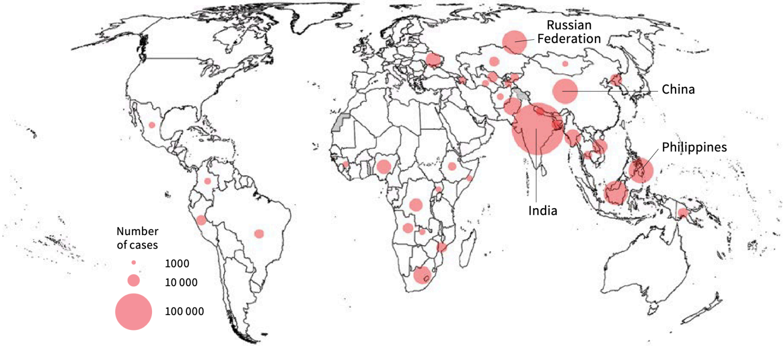


World Health Organization, 2025

Drug resistance cases

FIG. 8

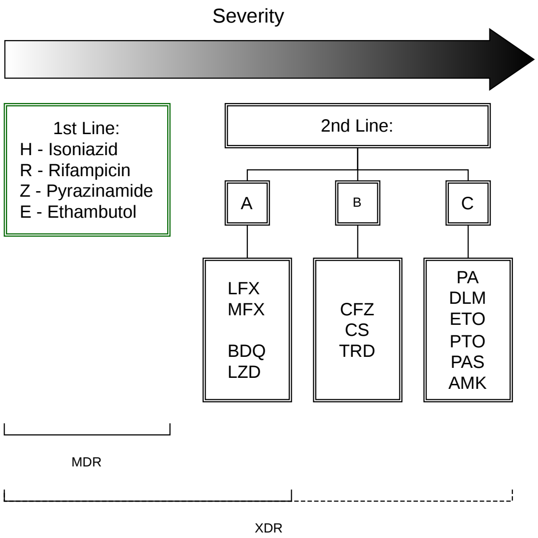
Estimated number of people who developed MDR/RR-TB (incident cases) for countries with at least 1000 incident cases, 2024[a]



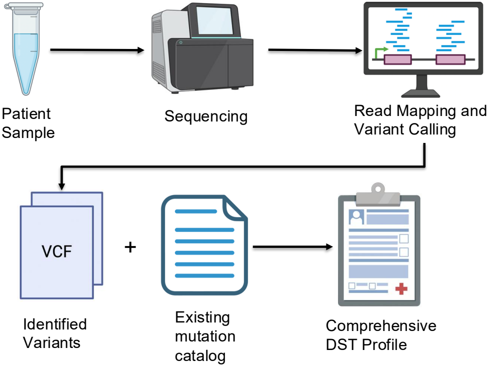
[a] The labels show the four countries that accounted for more than half of the global number of people estimated to have developed MDR/RR-TB in 2024.

World Health Organization, 2025

TB treatment regimen

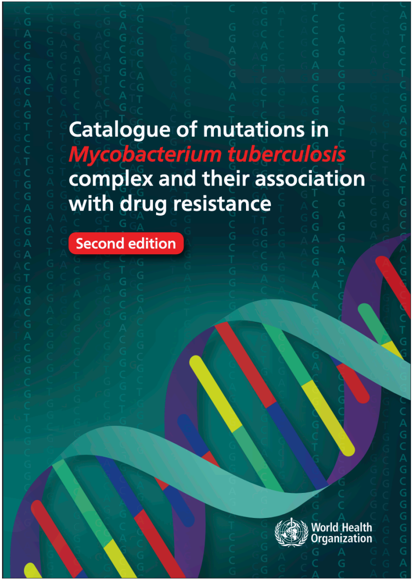


Current WGS pipeline



- ~ 7 – 14 days
- Comprehensive
- Fast
- Incomplete Coverage

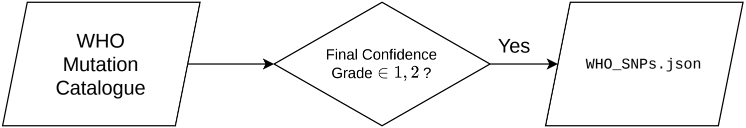
- International standard for interpreting genotypic data for DST
- Supports national TB programs and diagnostic developers – improves:
  - treatment decisions thus patient outcome



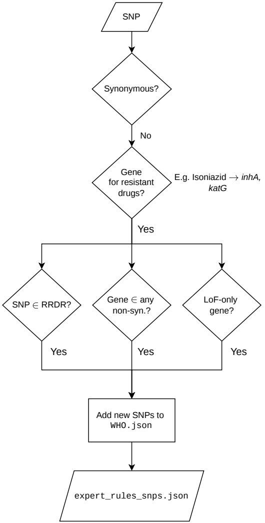
Aims

1. Identify gaps in the WHO mutation catalogue for drug resistance prediction
2. Identify candidate SNPs to address these gaps using WHO expert rules and feature ablation
3. Develop a neural network-based alternative to rule-based AMR prediction tools

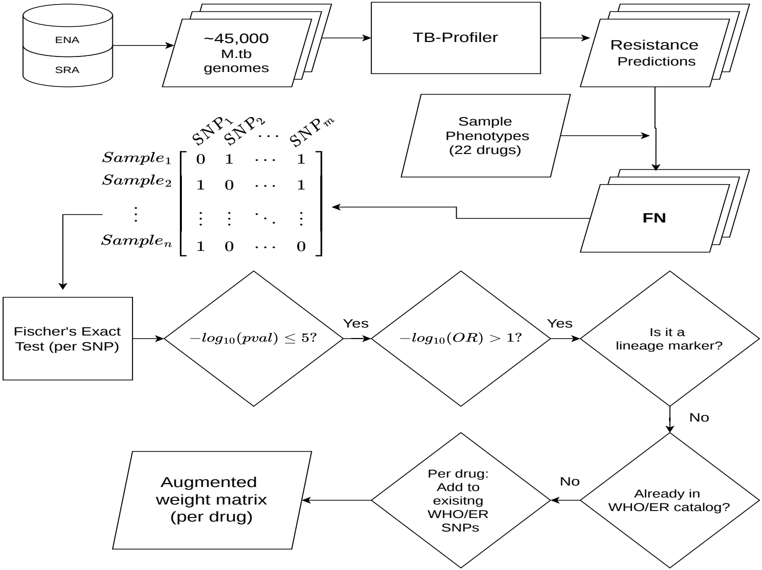
Converting the SNP catalogue for an NN



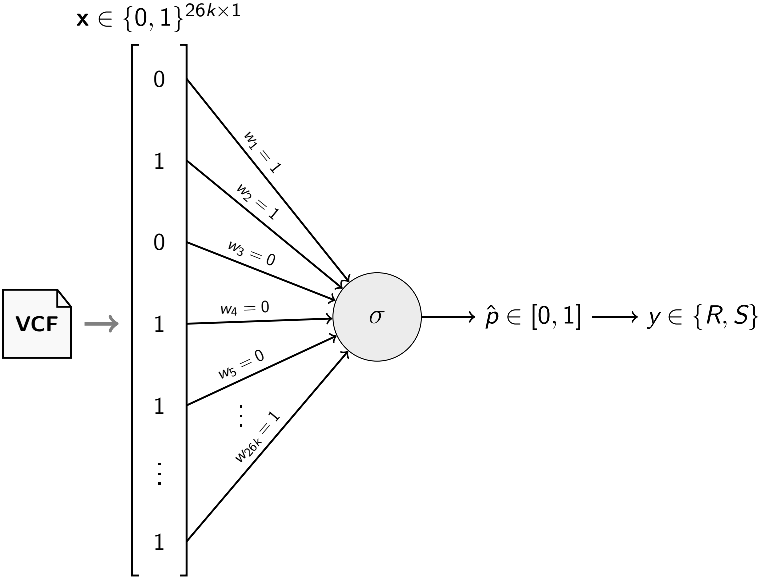
Updating the WHO catalogue with expert rules



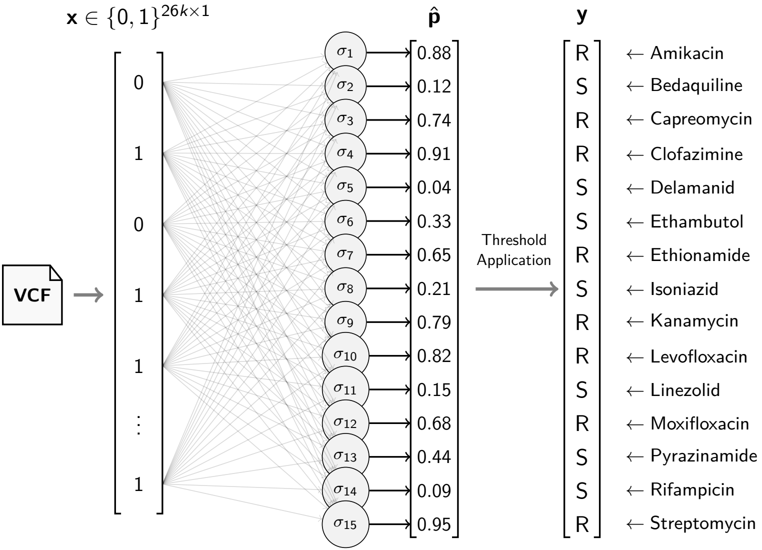
Augmenting our SNP lists from FN samples



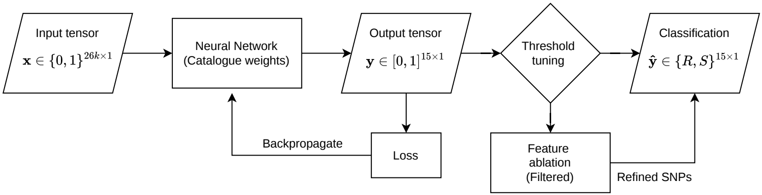
Schematic of a single neuron



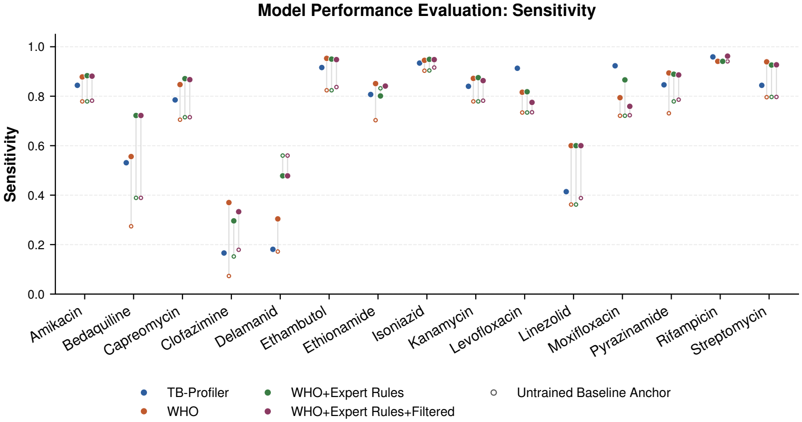
Schematic for all drugs from the WHO catalogue



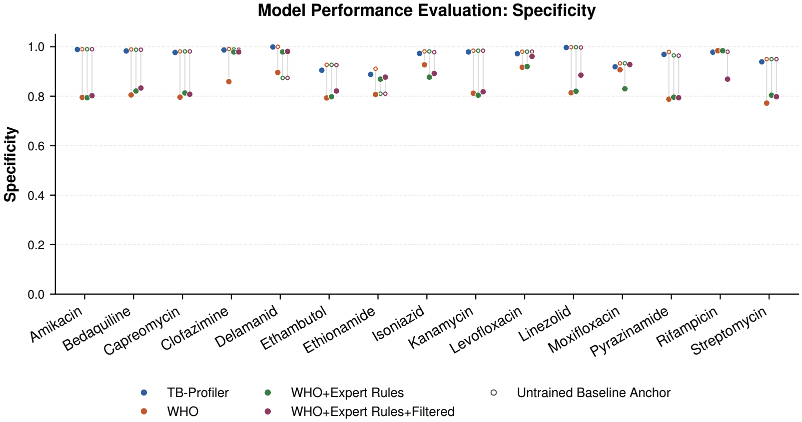
Training schematic



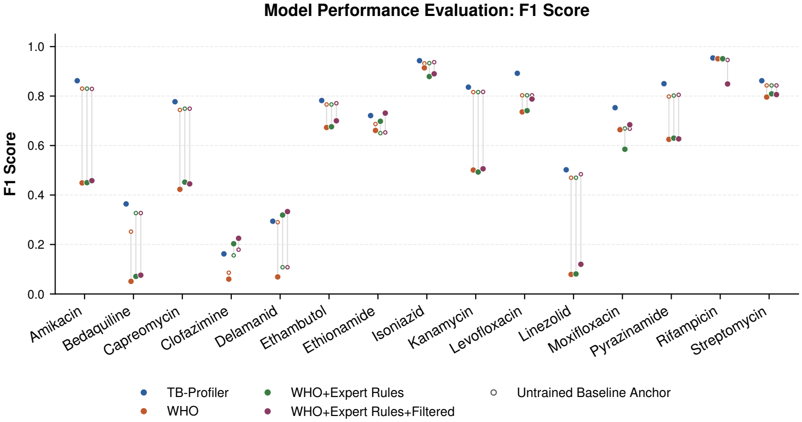
Diagnostic accuracy (Sensitivity)



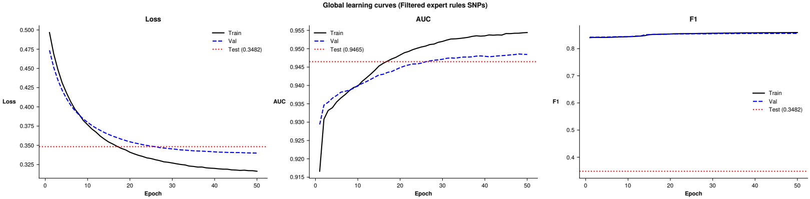
Diagnostic accuracy (Specificity)



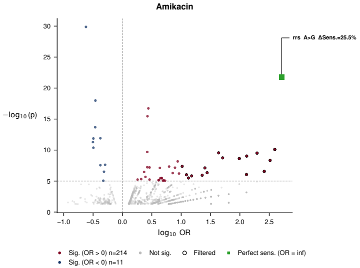
Diagnostic accuracy (F1 Score)

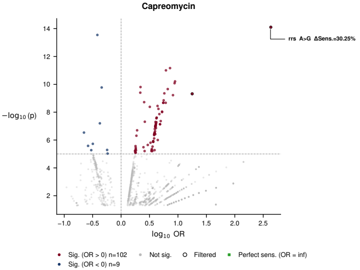


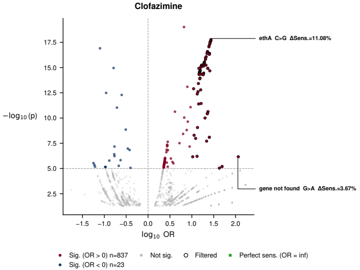
Learning performance for augmented WHO catalogue

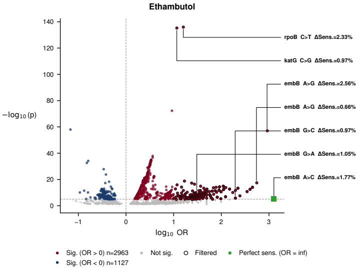


SNP ablation - changed sensitivity

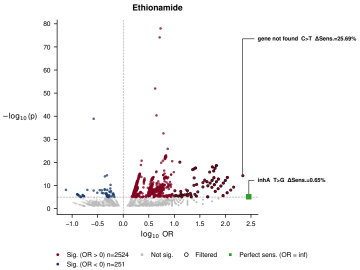


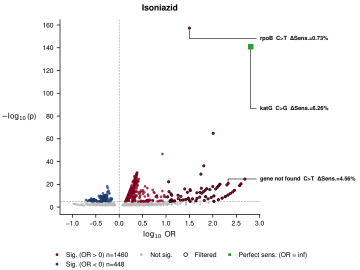


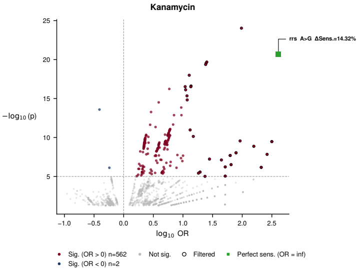




SNP ablation - changed sensitivity







Clinical interpretation

- High sensitivity prioritised over specificity because we aimed to reduce FNs
  - Lower FNs $\implies \uparrow$ TPR $\implies \downarrow$ Missed resistant cases
  - Lower Spec. $\implies \uparrow$ FPs $\implies$ Susceptible cases treated as resistant
- False negatives in drug resistance predictions may result in ineffective treatment
  - Patient administered drug that the M.tb sample is resistant to $\implies$ Poorer health outcomes
- Catalogue-weighted schemes outperformed or matched TB-Profiler (baseline) for 14/15 drugs

Ablation biological significance

- SNPs retained post-ablation are consistent with known resistance mechanisms
- Novel candidate SNPs have been identified in 3/4 first-line drugs

Limitations

- Training data lacks resistant cases for newer drugs
- Class imbalance in resistant sample for BDQ, LZD, DLM, CFZ
- Model validated on same data as training

Next steps

THE UNIVERSITY OF MELBOURNE

- Validation from external cohort (CRyPTIC consortium, MIRU-VNTR)
- Lineage stratification for evaluation
- Wet-lab validation of novel SNPs
- Implement different network architechture (XGBoost, hidden layers for shared mechanisms (inhA), etc.)
- Deployment study - measure real-world clinical outcomes vs rule-based AMR prediction tools (TB-Profiler)

References

- World Health Organization. Global tuberculosis report 2025. Geneva: World Health Organization; 2025. Available from: https://www.who.int/publications/i/item/9789240116924
- D'Souza, C., Kishore, U., & Tsolaki, A. G. (2023). The PE-PPE family of Mycobacterium tuberculosis: proteins in disguise. Immunobiology, 228(2), 152321.
- Hall, M. B., & Coin, L. J. (2022). Assessment of the 2021 WHO Mycobacterium tuberculosis drug resistance mutation catalogue on an independent dataset. The Lancet Microbe, 3(9), e645.
- Hall, M. B., Wick, R. R., Judd, L. M., Nguyen, A. N., Steinig, E. J., Xie, O., ... & Coin, L. (2024). Benchmarking reveals superiority of deep learning variant callers on bacterial nanopore sequence data. eLife, 13, RP98300.
- Walker, T. M., Miotto, P., Köser, C. U., Fowler, P. W., Knaggs, J., Iqbal, Z., ... & Paton, N. I. (2022). The 2021 WHO catalogue of Mycobacterium tuberculosis complex mutations associated with drug resistance: a genotypic analysis. The Lancet Microbe, 3(4), e265-e273.
- World Health Organization. (2022). WHO consolidated guidelines on tuberculosis. Module 4: treatment-drug-resistant tuberculosis treatment, 2022 update. World Health Organization.
- World Health Organization. (2023). Catalogue of mutations in Mycobacterium tuberculosis complex and their association with drug resistance (2nd ed.). World Health Organization. https://www.who.int/publications/i/item/9789240082410